



School of Computer Science and Engineering | Greater Noida Campus

B.Tech CSE Summer Internship Evaluation & Technical Project Defense

Full-Stack Web Application Development, n8n Automation & Cloud Deployment

Engineering reactive edge web platforms, serverless backend microservices, and asynchronous event workflows

CANDIDATE INFORMATION

Student Name: Aman Kumar Jha

Enrollment / Roll No: 2410030417

Program & Branch: B.Tech Computer Science & Engineering

Semester & Section: 5th Semester | Section 3CSE7

Academic Batch: Batch 2024–28

ORGANIZATION & ROLE

Host Organization: Zeta Galaxy

Industry Domain: AI Solutions & Full-Stack Automation

Official Designation: Software & Automation Engineer Intern

Internship Tenure: 1st June 2026 – 30th July 2026 (Summer 2026)

Host Location: Samridhi Grand Avenue, Greater Noida, U.P.

Executive Presentation Agenda

01

Organization Profile

Zeta Galaxy domain, technology ecosystem & intern role

02

Problem Statement

Industry challenges in deployments, latency & manual sync

03

Project Objectives

Full-stack milestones, n8n workflows & cloud hosting

04

Tech Stack Architecture

Next.js 14, TypeScript, Node.js, n8n, PostgreSQL, Vercel

05

System Architecture

Multi-tier decoupled edge design and serverless functions

06

n8n Automation Engine

Event webhooks, asynchronous data pipelines & triggers

07

CI/CD & Cloud Delivery

Automated GitHub Actions, preview URLs & Vercel CDN

08

Key Outcomes & Impact

Lighthouse 96+, sub-90s deploy time & core learnings

Organization Profile: Zeta Galaxy

ABOUT ZETA GALAXY

- **Company Profile:** High-growth technology enterprise delivering modern web applications, AI-driven solutions, and automated cloud systems.
- **Headquarters:** Samridhi Grand Avenue, Greater Noida, Uttar Pradesh.
- **Core Engineering Focus:** Cloud-native architectures, low-code/node automation, reactive web applications, and serverless edge compute.
- **Engineering Standards:** Agile Scrum sprints, GitOps continuous delivery, automated unit/integration testing, and zero-downtime releases.

ROLE & CORE RESPONSIBILITIES

- **Full-Stack Development:** Architecting responsive Next.js/React frontend interfaces with strict TypeScript typing and Tailwind styling.
- **Serverless Backend APIs:** Constructing RESTful API endpoints, JWT token authentication middleware, and Prisma ORM database layers.
- **n8n Workflow Automation:** Designing event-driven n8n automation pipelines for webhook listening, asynchronous data transformation, and alerts.
- **Continuous Cloud Delivery:** Managing CI/CD workflows via GitHub Actions and automated deployments to Vercel's global edge network.

Problem Statement & Industry Challenges

01. Manual Deployment Friction

Deploying software via manual SSH sessions and unversioned server scripts caused release downtime, human error, and staging configuration drift.

02. Repetitive Operational Overhead

Manual data synchronization, customer event dispatching, and notification routing consumed valuable engineering bandwidth and caused delays.

03. Monolithic Latency & Scaling Limits

Centralized server monoliths suffered from high latency for distributed users and lacked rapid autoscaling during traffic spikes.

04. Fragmented Quality Assurance

Absence of automated pull-request validation allowed untested code and regression bugs into production, elevating maintenance costs.

Project Objectives & Technical Scope

1. Modular Full-Stack Architecture: Build responsive, accessible user interfaces in Next.js 14 & React 18, paired with type-safe TypeScript interfaces and Tailwind CSS.

2. Serverless Backend & API Layer: Develop structured REST API endpoints, JWT token authentication, and type-safe database queries via Prisma ORM.

3. n8n Workflow Automation: Orchestrate asynchronous webhook listeners, automated data transformations, and cross-platform notification pipelines.

4. Automated CI/CD Pipeline on Vercel: Establish zero-downtime deployment pipelines with GitHub Actions and Vercel CLI for automated branch preview URLs and production rollouts.

5. Performance & Core Web Vitals: Optimize bundle size, leverage dynamic code splitting, and achieve 95+ Google Lighthouse scores across Core Web Vitals.

Technology Stack & Tooling Architecture

Frontend UI & UX

- Next.js 14 (App Router & SSR)
- React 18 & TypeScript
- Tailwind CSS & Lucide Icons
- Client / Server Components

Backend & APIs

- Node.js & Route Handlers
- RESTful API Endpoints
- JWT Authentication Middleware
- Zod Data Validation

Workflow Automation

- n8n Workflow Engine
- Event Webhook Listeners
- REST API Connector Nodes
- Asynchronous Pipelines

Database & ORM

- PostgreSQL Database
- Prisma ORM Query Builder
- Database Connection Pooling
- Automated Schema Migrations

Cloud & Deployment

- Vercel Global Edge Network
- Serverless Edge Functions
- Custom Domain Routing
- Edge Caching & ISR

DevOps & Testing

- GitHub Actions Workflows
- Vercel CLI & Bot Triggers
- Jest & React Testing Library
- ESLint & Prettier Standards

System Architecture: Decoupled Edge Multi-Tier Design

1. Presentation & Edge Layer [Next.js App Router | Vercel Global Edge CDN]

Routes static assets with sub-50ms latency; executes Edge Middleware for session routing.

2. Serverless API Layer [Node.js Serverless Functions | REST Endpoints]

Stateless execution on-demand; scales automatically from 0 to thousands of instances.

3. n8n Automation Engine [Event Webhooks | Workflow Pipelines]

Asynchronously handles event dispatches, multi-step business logic, notifications & third-party sync.

4. Data Persistence Layer [PostgreSQL Cloud Database | Prisma ORM]

Type-safe relational storage with connection pooling, transactional consistency, and ACID guarantees.

n8n Workflow Automation Implementation

- **Webhook Trigger Node**

Listens for real-time application events (user actions, form submissions, state changes) via secure HTTP webhooks.

- **Data Transformation & JSON Parsing**

Extracts, validates, and normalizes payloads using JavaScript Code nodes and built-in n8n expressions.

- **Multi-Service Orchestration**

Coordinates actions across databases, notification services (Slack, Email, SMS), and external cloud APIs without server overhead.

- **Error Catching & Logging**

Configured automated error-handling workflows to capture failed executions and send instant incident alerts to engineering channels.

CI/CD & Cloud Deployment on Vercel

Step 1: Local Development

Developers write code with TypeScript & Tailwind CSS; pre-commit hooks verify linting and formatting.

Step 2: GitHub Actions CI

Pushing to GitHub triggers automated test suites (Jest) and type checks to block regression errors.

Step 3: Preview Deployments

Every pull request generates an immutable preview URL for team review and end-to-end testing.

Step 4: Production Release

Merging to the main branch triggers an atomic zero-downtime build deployed across Vercel's global CDN in < 90 seconds.

Key Results & Quantitative Impact

< 90s

Automated Build Time

Continuous deployment to Vercel completed in under 90 seconds from merge.

96+

Lighthouse Performance

Exceptional Google Lighthouse Core Web Vitals score across all pages.

100%

Zero Manual Release

Eliminated manual SSH deployments; 100% automated GitOps workflow.

> 85%

Automated Test Coverage

High test coverage across critical backend routes and core UI components.

Key Learnings & Professional Growth

- **Industry Full-Stack Engineering**

Mastered server-side rendering, component-level state management, and edge middleware routing in production Next.js applications.

- **Event-Driven Architecture**

Learned to decouple business logic from background tasks using n8n webhook pipelines, increasing application responsiveness.

- **Production Cloud Delivery**

Gained deep practical expertise in Vercel edge deployment, environment variable management, and GitOps workflows.

- **Agile Team Collaboration**

Actively contributed to sprint planning, standups, code reviews, and issue tracking in a collaborative engineering culture.

Thank You!

Questions & Technical Discussion

Aman Kumar Jha | Roll No: 2410030417 | B.Tech CSE (Batch 2024–28, Section 3CSE7)
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